



FLIP THROUGH BEFORE YOU DOWNLOAD

Never Type a Total

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RULE 1

Compute it from the parts, every time

The rule in one line: **if a number can be worked out from other numbers on the page, you do not type it.** Not into a document, not into a spreadsheet cell, not onto a poster.

Why proofreading does not catch it, and this is the whole problem

A wrong total looks approximately right. \$80.44 is in the neighborhood of \$84.40. The eye checks the shape of a number, not its value, and a transposed digit keeps the shape.

And nobody re-checks a number they typed themselves. You worked it out once, carefully, and from then on it is furniture. The second read is reading your own earlier confidence.

The part that actually bites, months later

The typo is not the real danger. **The real danger is the change.** A price goes up, a question gets dropped, a rubric row is reworded and reweighted. The parts move and **the typed total stays exactly where it was, now confidently wrong,** and nothing on the page indicates that anything happened.

What this looks like with no software at all

In a spreadsheet: the total cell never has a number. If you have ever typed a figure into a cell under a column of numbers, that is the problem this page is about.

In a document: if the total cannot be a formula, **do not print it.** A materials list with no total is correct forever. A materials list with a stale total is worse than useless, because people trust it.

On a rubric: state the weights, and let the points be the sum of the rows. The "out of 100" at the top is the single most commonly wrong number in a teacher's documents, and it is wrong because somebody reweighted a row in August.

Build it so the wrong thing is impossible, rather than so that you are careful.

This is the same move as not having a box to type a student's name into, and the same move as giving every student a differently ordered test instead of policing copying. **You do not remember a rule. You remove the way to break it.**

SNEAK PEEK · 5 pages

RULE 2

Keep money in whole cents

Here is the part that catches people who did everything else right. **Computers do not store decimal money exactly**, so even a computed total can come out wrong-looking. These are real outputs, not an illustration.

WHAT A COMPUTER ACTUALLY RETURNS

$0.1 + 0.2$	→	0.30000000000000004
8.44×10	→	84.39999999999999
1.10×3	→	3.3000000000000003
$2.35 + 1.10 + 0.95 + 4.04$	→	8.440000000000001

Why this matters even though it is a tiny amount

Nobody loses money over four zeros and a one. **What you lose is trust**, in front of a class, when the board says \$8.440000000000001 and thirty students decide the software is broken. And then you round it for display, you have two different numbers in the system and no rule about which one to use.

The fix, and it is one so

Store money as a whole number of cents and divide only when you show it. \$8.44 is 844. Ten of them is 8,440. Shown, that is \$84.40, exactly, every time.

The same four parts above, added as cents and divided at the end, give **8.44** with nothing after it. The arithmetic did not change. **What changed is that the rounding happens once, at the end, where you can see it.**

And the one that caught me: fractions typed as decimals

A shop runs on fractions, and so does a recipe, a sewing pattern and a scale drawing. **A board is three-quarters of an inch, not 0.75**, and the moment somebody retypes it as 0.75 you have lost the thing the student was supposed to read. Keep the fraction the student sees, **parse it for the arithmetic**, and never make a child read a decimal of an inch because the software found it easier.

If you are not writing software, the same rule reads:

round once, at the end, where somebody can see it happen. Not in the middle of a column, not differently in two places, and never twice.

Six places to look, and what to do about each

Go through the materials you already use. **You are looking for numbers you typed that something else could work out.** This is quicker than it sounds and it rarely comes up empty.

#	Where to look	What to do
1	Points possible at the top of a test or a rubric	Make it the sum of the rows, or delete it.
2	Answer keys where an answer is arithmetic on the question	Work every one again from the question, not from the key.
3	Materials and supply lists with a cost at the bottom	A formula, or no total at all.
4	Letters home quoting a price or a date count	Check against the source the day you send it.
5	Anything you copied from last year	Every number in it is a year old. Treat it as unverifiable.
6	A total that appears twice in one document	Delete one of them. Two copies of a number is two chances to be wrong.

SNEAK PEEK · 5 pages

The one nobody expects to find

Row 6. A figure in the body and the same figure in a summary box is the most reliable source of contradiction in any document, because only one of them ever gets updated. **If you find a document where the two disagree, you have found out which one people actually read.**

And if you ask AI to help with any of this

It will happily add a column for you and it is often right. **It is also perfectly willing to produce a total that is confidently wrong**, in the same calm voice, and it will not tell you which kind of answer you just got. **Ask it for the formula rather than the answer.** A formula you can check in two seconds; a number you have to redo by hand.



WHAT IS INSIDE

Never Type a Total

- ✓ The total is never typed - it is computed from the parts
- ✓ Why proofreading does not catch a wrong total
- ✓ Keep money in whole cents, with the real outputs shown
- ✓ Fractions typed as decimals, and what it costs a student
- ✓ Six places to look in materials you already use
- ✓ A fill-in page, so you leave having fixed one

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